Ankita Sharma

7006810587 ankitasharma102002@gmail.com Ankita Sharma ankita-1007 Portfolio

Education

Thapar Institute of Engineering and Technology, Patiala **Sep 2020 – Jun 2024**
Bachelor of Engineering in Computer Science Engineering *CGPA: 8.75/10*

Army Public School Jammu Cantt, Jammu **Mar 2019 – Apr 2020**
CBSE XII (Senior Secondary) Science *95.6%*

Work Experience

Airtel Digital **Jul 2024 – Present**
Software Developer

▷ **Backend Engineering** | *Java · Spring Boot · REST*

- Developed Java Spring Boot microservices and REST APIs powering the MITRA Airtel retailer application — a key revenue-generating platform.
- Developed a DoT-mandated compliance portal for J&K and NESA telecom circles — built paginated REST APIs for document upload, bulk-upload, approval workflows for police verification documents, and real-time DB retrievals for SIM-selling retailer status, with automated restriction enforcement based on document expiry.
- Built key features for RAAHI, an internal Airtel employee travel reimbursement app — implemented trip pause/resume functionality, dynamic rerouting to manager approval flows, conditional skip-manager logic for edge cases, and distance validation for reimbursement eligibility.
- Enhanced customer retention on DTH orders by integrating geolocation data into backend order processing, and improved system security through login cache optimization and session invalidation.
- Managed project workflows using JIRA and Bitbucket for tracking, code reviews, and collaboration.

▷ **Data Science** | *Python · YOLOv8 · OpenCV*

- Trained YOLOv5 and YOLOv8 models for real-time object detection, improving inference accuracy in production pipelines.
- Investigated facial recognition technologies for KYC processes, strengthening user verification and security.

IIT Jammu **Jan 2024 – Jun 2024**
Research Intern | Deep Learning & Network Security

- Built ML models using transformers and autoencoders to detect PII leakage in app network traffic; applied triplet loss to reduce overfitting.
- Leveraged GANs to synthetically generate training data, improving model robustness and testing coverage.

Microsoft **May 2022 – Jun 2022**
Intern Engage Mentee '22

- Built an ATM security system using facial recognition for password-free login via OpenCV and Tkinter. Git

Technical Skills

Concepts: Data Structures, Algorithms, DBMS, OOP, Microservices, Distributed Systems, Event-driven Architecture, Concurrency
Languages: Java, Python, C/C++, SQL, R (familiar)
Frameworks: Spring Boot, Spring Data JPA, Hibernate, Flask (familiar)
Messaging & Cache: Apache Kafka, Redis
Libraries: pandas, NumPy, Matplotlib, OpenCV, scikit-learn, Keras
Cloud & DevOps: Docker, Git, Firebase
Tools: Jira, IntelliJ, Bitbucket, Postman, Jupyter Notebook, Tableau

Projects

Event Booking System | *Java 21, Spring Boot, Redis, PostgreSQL, Docker* | *Git* **Jul 2026**

- Built a scalable event booking system with microservices architecture — separate Event Booking and Notification services communicating via REST.
- Implemented distributed seat locking using Redis atomic operations (SETNX + TTL), preventing double-booking under concurrency with automatic lock expiry and user-specific lock refresh.
- Designed end-to-end booking workflow — seat locking, confirmation, PostgreSQL persistence, and email notification via Notification Service with idempotency support.

Notification System | *Java, Spring Boot, Kafka, PostgreSQL, Firebase, Docker, Groq AI* | *Git* **Jun 2026**

- Built an asynchronous, fault-tolerant notification system using Kafka, decoupling request handling from multi-channel delivery (Email via SMTP, Push via Firebase).
- Designed a 3-tier retry mechanism with increasing backoff and a Dead Letter Queue for exhausted failures, with idempotency to prevent duplicate processing.
- Integrated an LLM (Groq/Llama 3.3) to auto-generate plain-English failure analysis for DLQ entries and perform fail-open spam/abuse detection on incoming requests.

TOPSIS - Multiple Criteria Decision Making | *Python, Pandas, NumPy, Flask* | *Git* **Jan 2023**

- Implemented the TOPSIS algorithm with a Flask web interface for multi-criteria decision making.
- Published as a pip-installable package on PyPI. PyPI package